

Attention: Faisal D'Souza, NCO, 2415 Eisenhower Avenue, Alexandria, VA 22314, USA

Gordon Broadbent and I, Newman Cheng, are humbled to submit this response to OSTP's RFI on behalf of NITRD NCO regarding the Development of an Artificial Intelligence Action Plan. We bring a shared commitment from the combined lens across elements of government, industry, and academia to build a comprehensive roadmap that will ensure America's continued success in AI innovation. We appreciate the time and for considering our submission. For any direct questions or further information, please do not hesitate to reach out to us.

Very Respectfully,

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An Imperative Strategic Shift: Acknowledging a Roadmap Deficit

With the People Republic of China's (PRC) 2027 and 2035 military modernization milestones on the horizon, America's perceived strategic advantage is a precarious illusion.¹ The PRC's ability to eventually synergize advancements in AI with domains such as cyber, autonomy, and science will equip them with the capabilities and influence to transform the existing world order, an objective they are committed to realizing.² The United States (U.S.) has been riding a wave of technological complacency, not due to a lack of innovation but because groundbreaking advancements are failing to translate into national strategic capabilities. Vannevar Bush's words still ring true, "A nation which depends upon others for its new basic scientific knowledge will be slow in its industrial progress and weak in its competitive position in world trade." But despite significant investments and numerous technological breakthroughs, there is a disconnect between innovation and its effective integration into the strategic national capabilities that bolster integrated deterrence. Innovation fueled by government investments is a challenge well addressed in today's ecosystem; the National Science Foundation continues to lead in driving critical investments. Investments alone do not translate into tailored capabilities for the U.S. to capitalize on. The real challenge comes in integration. In former Google Chief Executive Officer (CEO) Dr. Eric Schmidt's comments to the House Armed Services Committee (HASC) in 2018, he stated that "the DoD does not have an innovation problem. It has an innovation adoption problem." The crux of the issue is that "while change has inherent risk, ... there is more at stake by maintaining the status quo in the face of quickly-evolving security threats and adversaries that are accelerating their technological advancement... with this in mind, everyone in the world - including US competitors - has equal access to the technology."³ The innovation adoption bottleneck persists and is only exacerbated by the increasing speed and impact of new AI innovations. This gap has resulted in missed opportunities to harness these advancements for maintaining and enhancing U.S. strategic dominance, particularly in the face of rapid progress by global competitors like the PRC.

¹ Page 4, 34, [U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China, 2023](#)

² Page 10, 11, 30, 32, [Annual Threat Assessment of the U.S. Intelligence Community](#)

³ [Statement of Dr. Eric Schmidt House Armed Services Committee April 17, 2018](#)

The rapid and comprehensive modernization of China's military directly challenges U.S. and allied dominance in the Indo-Pacific region.⁴ The PRC increasingly views warfare as a confrontation between opposing operational systems rather than annihilation between opposing military forces that are mechanical in nature. This perspective has shifted the PRC's transformation goals from largely mechanical in focus towards an information dominance one, enabling them to fight intelligence-fueled warfare effectively.⁵ Thus, significant investments directed by the Chinese Communist Party (CCP) have been funneled into PRC industry as well as their R&D sector to support this goal.^{6,7} From 2014 to 2023, PRC inventors filed 38,000 patents in the generative AI domain, with the Chinese Academy of Sciences contributing 1,100 of those. In stark contrast, top U.S. industry and science institutions collectively filed only 6,300 patents in the same period, with Stanford University accounting for 600 of those.⁸ We have an urgent need to not only maintain an industrial ecosystem that can drive innovation, but also establish new avenues for government to integrate them more seamlessly into tailored capabilities that can bolster a strategic advantage.

Harnessing Knowledge for National Strength

To secure victory in the current era of great power competition, the United States (U.S.) requires a robust and comprehensive strategy that clearly defines the role of Artificial Intelligence (AI) in strategic position. This strategy must not only operationalize our capabilities against our adversaries but also integrate AI technologies that abide by U.S. and allied values. In the current geopolitical landscape, winning great power competition entails continually outpacing adversaries and applying tailored instruments of national power to elicit behavioral changes that advance any nations' core interests.⁹ Great power competition has traditionally been measured through a blend of hard and soft power through military strength, economic leverage, and diplomatic influence. Today, however, that landscape has shifted; knowledge has emerged as a critical indicator of national power.¹⁰ The generation of new knowledge and how it is applied can enhance both hard and soft power while restricting another nation's ability to develop such knowledge provides a strategic technological advantage.

In any competitive domain, the key to victory for any person, organization, or nation lies in generating, applying, and controlling knowledge faster and more effectively than their adversary. This requires more than just data lakes; it requires data to be part of a system that creates actionable, useful knowledge. AI is key to transforming information stored within unmanageable datasets into knowledge. This principle holds true in great power competition, especially in the 21st century where the world is immensely more complex than any prior decade. The vast array of variables that shape the competitive landscape often leads to situations where the best course of action may be inaction. In this intricate environment, ongoing integrated deterrence is crucial to prevent great power competition from escalating into full-scale conflict. Integrated deterrence does not have an end state; it is a living challenge that requires continual iteration and adaptation in order to prevent spiraling into war. While the U.S. stands ready to address such conflicts, if necessary, leveraging

⁴ VII, Page 18, [U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China, 2023](#)

⁵ Page 40, 97 [U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China, 2023](#)

⁶ Page 9, [Annual Threat Assessment of the U.S. Intelligence Community](#)

⁷ Page 21, 33, 169, 166 [U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China, 2023](#)

⁸ Page 9, [Generative Artificial Intelligence, Patent Landscape Report, 2024](#)

⁹ Page 22, [National Security Strategy, October 2022](#)

¹⁰ [The Crumbling Foundations of American Strength](#)

AI to achieve effective integrated deterrence offers significant advantages. Our key adversaries in the 21st century have both adopted national security strategies that reflect the concept of integrated deterrence (integrated strategic deterrence and *sderzhivanie*), and show a clear intent to fight and gain the advantage in operating within a multi-domain environment using AI^{11, 12}. Each of America's adversaries are different, and to out-compete the PRC and contain Russia, winning integrated deterrence means that the U.S. needs a comprehensive AI roadmap applied to government, industry, and scientific discovery.¹³ The AI roadmap we lay out is much broader than just military application; we focus on how to keep the U.S. competitive edge. Our AI roadmap is a comprehensive strategy designed to ensure the U.S. maintains its competitive edge across various sectors. The focus is on maintaining a holistic national dominance in the economy (industry), scientific discovery, and national defense. The perpetual need for adaptation through integration highlights the critical role of knowledge and innovation, an area where the U.S. has had a longstanding competitive advantage. To succeed in great power competition, nations must adeptly apply and manage a healthy, AI-integrated ecosystem across the spheres of government, science, and industry to build an ongoing competitive advantage for them and their allied partners.

The Role of Government, Science, and Industry

In the context of great power competition, the traditional roles of government, science, and industry must be complementary and interdependent, each contributing to the broader national interest through distinct yet interconnected functions. Historically, the government's role has centered on providing direction and resources by paying for the fundamental research, development and production of capabilities, particularly during critical periods such as World War II (WWII) and the Cold War. During these times, the U.S. government acted as the primary financier and organizer of large-scale research and development initiatives that were instrumental in establishing the U.S. as a technological and military superpower. In this traditional framework, advances in science have served as the engine of innovation, fueled by direct government investment driving the creation of new knowledge and technologies. The academic and research institutions, often funded by government grants, were pivotal in advancing the foundational theories and discoveries that would be harnessed for national defense and other strategic purposes. This drove advances in technologies that had military or national security applications as their priority (i.e., stealth technologies and GPS), but then led to commercial applications as well.

The symbiotic relationship between government and science helped ensure that cutting-edge research was aligned with national priorities, leading to breakthroughs that were not only scientifically significant but also strategically advantageous. On the other hand, industry has been the primary agent of applying and scaling these scientific advancements into practical, operational technologies. The private sector's role in manufacturing, technology development, and commercialization has been crucial in translating scientific discoveries into tangible assets that can be deployed nationally and globally. The defense industrial base (DIB), driven by direct government requirements and funding, exemplifies this dynamic, where industry partners followed the government funding and led in developing and producing the sophisticated technologies that

¹¹ Page 96, [U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China, 2023](#)

¹² Page 12, [Integrated Deterrence as a Defense Planning Concept](#)

¹³ Page 23, [National Security Strategy, October 2022](#)

defined the 20th century. In this traditional model, innovation, and technology advancements focused on national security followed a relatively linear path: the government set the requirements and provided funding; the government invested in scientific discovery focused on generating the necessary knowledge; and industry brought that knowledge to the market or battlefield. Lockheed Martin's F-35 Lightning II is a clear example of the U.S. Government's requirements and desires to possess a fifth-generation stealth fighter. Fundamental science needed to be pushed through innovations to enable the development of a fifth-generation stealth fighter. Lockheed Martin then brought them to market for the U.S. and now additional countries as well.¹⁴

However, this paradigm is evolving in the 21st century, particularly with the rise of AI and other advanced technologies. Science, in particular national labs and academia, continues to play a critical role in foundational research, but it must also adapt to the faster pace of technological development driven by industry. In this environment, Former Secretary Mattis's words particularly ring true, "Success no longer goes to the country that develops a new fighting technology first, but rather to the one that better integrates it and adapts its way of fighting...Our response will be to prioritize speed of delivery, continuous adaptation, and frequent modular upgrades. We must not accept cumbersome approval chains, wasteful applications of resources in uncompetitive space, or overly risk-averse thinking that impedes change."¹⁵ The roles between industry and government are becoming more fluid, with the industry now driving much of the technological innovation and the government struggling to adopt innovate, particularly in information technology, which is surpassing the pace of government-funded integration, adoption, and bureaucracy. This shift necessitates a reevaluation of how these sectors interact. While still essential in setting policy and strategic direction, the government must now focus on creating an environment that fosters rapid innovation and ensures national security without stifling the dynamism of the private sector. Meanwhile, the information technology industry has assumed a more prominent role as the primary innovator. For example, the Magnificent 7 Stocks (MAG7) are leading in the deployment of AI and infrastructure investments to train more powerful AI models.¹⁶ Today, there is an inherent symbiotic relationship between the MAG7 companies and small and medium sized businesses (SMBs); OpenAI would not exist without Microsoft's infrastructure to train the GPT-series models and DeepMind would not exist without Google's infrastructure. Without the MAG7's money and GPUs, fundamental research breakthroughs emerging from industry would not be possible. To maintain a competitive edge in knowledge creation in an environment where large technology companies are leading in advancements, the traditional operating model between government, science, and industry needs to change. The U.S. and its allies should focus less on developing capabilities and more on: fostering the ecosystem, leveraging the investments already made, and integrating the advances to maintain the U.S.'s competitive edge in great power competition.

Government's Evolving Role in a Techno-Centric 21st Century

During WWII and the Cold War, centralized high-tech research efforts funded directly by the government demonstrated the decisive role of technological innovation at home in shaping

¹⁴ [F-35 Brochure_3-2020.pdf \(lockheedmartin.com\)](#)

¹⁵ [Remarks by Secretary Mattis on the National Defense Strategy > U.S. Department of Defense > Transcript](#)

¹⁶ Page 36, [Generative Artificial Intelligence, Patent Landscape Report, 2024](#)

global outcomes. The government's combined investments into the Manhattan Project and Aerospace Race gave birth to a crucial defense industrial base and specialized research shops, such as Skunk Works at Lockheed Martin and Phantom Works at Boeing. Over the five-year duration of the Manhattan Project, the total investment by the U.S. government, adjusted for today's inflation, amounted to approximately \$32 billion¹⁷; a number overshadowed by Google's more than \$30 billion dollar investment into AI capabilities each year for the past two years. In 2024, Google is projected to invest more than \$50 billion into AI capabilities, exemplifying the scale of capital investments that industry is making into accelerating technology development and innovation. Google's investment is reflective of only one of the MAG7 companies.¹⁸ The others are making similar investments, showing that upfront capital investments into major technology initiatives and programs are fundamentally industry driven. The information technology industry is increasingly investing in AI to advance technologies that are and will continue to be critical to national security.

Diverse Minds Leads to Innovative Innovation

The Manhattan Project is a prime example of how diverse minds can lead to innovative innovation. During this monumental effort, a key engineering decision was made on which detonation pattern would work for the bombs when utilized. In the end, Little Boy and Fat Man used different approaches; Little Boy fired a piece of Uranium 235 into the main payload, whereas Fat Man employed an implosion-based technique.¹⁹ Furthermore, there were debates regarding uranium enrichment and plutonium production itself; the Oak Ridge facility focused on uranium enrichment using gaseous diffusion and electromagnetic separation, while the Hanford Site produced plutonium in nuclear reactors. In the grand picture, the Manhattan Project was immensely successful and largely able to overcome engineering challenges by fostering a diverse group of scientists and allowing them to pursue different paths simultaneously, the Manhattan Project overcame numerous technical challenges and uncertainties. Such an approach underscores the importance of diverse perspectives in driving innovation and solving complex problems, one that parallels today's development of continuously more powerful AI models.

Today, AI models are developed within a rich ecosystem across industry, academia, and national laboratories. For instance, OpenAI's models on Azure are at the forefront, with the new o1 model excelling at solving complex reasoning problems. Additionally, a variety of models from other industry leaders like Meta²⁰, Cohere²¹, and Mistral²², catering to specific needs across different sectors are important for accessibility to drive innovation.²³ The combined ecosystem ensures that AI can be tailored to and foster positive innovation in a manner reminiscent of the Manhattan Project's diverse approach.

¹⁷ [The Manhattan Project: The race to build the atomic bomb - Bulletin of the Atomic Scientists](#)

¹⁸ [Magnificent 7 Stocks Earnings Updates: AI Remains the Focus](#)

¹⁹ [The Manhattan Project](#)

²⁰ [Introducing Meta Llama 3: The most capable openly available LLM to date](#)

²¹ [Cohere](#)

²² [\[2310.06825\] Mistral 7B \(arxiv.org\)](#)

²³ [Azure Machine Learning Model Catalog and Collections](#)

The Growing Coupling between Industry and Great Power Competition

Economic strategies have historically served as potent instruments of deterrence. The MAG7, which contains the most valuable companies in the world, plays a crucial role in AI infrastructure, development, and research, significantly shaping geopolitical outcomes. The 2022 Ukraine-Russia conflict exemplifies the crucial role of U.S. companies in directly supporting and influencing a defense of democratic ideas, particularly in the realm of digital infrastructure and connectivity. For example, Microsoft has played a key role in safeguarding Ukraine's digital assets by rapidly migrating critical data and networks to a secure Azure baseline, thereby protecting them from ongoing cyberattacks and ensuring continuity.^{24,25,26} In addition, SpaceX's Starlink has provided essential satellite-based internet connectivity, enabling resilient communication channels for both civilian and military operations in Ukraine, even in areas where traditional infrastructure has been compromised.^{27,28,29} The influence of industry on the geopolitical stage today has the potential to alter the outcomes of global conflicts. Industry is inevitably entangled in geopolitical competition, as evidenced by NVIDIA's dominance in the graphics processing unit (GPU) market and its implications for the People's Republic of China's (PRC) reunification ambitions with Taiwan, stemming from Taiwan's critical role in the global semiconductor supply chain. Taiwan Semiconductor Manufacturing Company (TSMC), based in Taiwan, is a key player in producing advanced semiconductor chips, including those used by NVIDIA for their GPUs. NVIDIA relies heavily on TSMC to produce its high-end GPUs, which are essential for AI applications, and if the PRC were to move towards reunification with Taiwan, it would disrupt the supply of these critical components, impacting not only NVIDIA but also the broader tech industry that depends on them.^{30,31} This potential disruption places NVIDIA at the center of geopolitical tensions, as control over semiconductor manufacturing becomes a strategic asset. While TSMC's role in the semiconductor supply chain underscores the geopolitical significance of hardware, the influence of foundational AI technologies, developed by companies like Google and Microsoft, is equally profound.

Google played a pivotal role in the AI revolution, particularly in the development of large language models (LLMs) and the foundational technologies behind them. In 2017, researchers from Google introduced the Transformer model in the paper "Attention is All You Need," which fundamentally transformed the field of natural language processing (NLP).³² The Transformer architecture, which relies on self-attention mechanisms, enabled more efficient processing of sequential data and significantly improved the ability to capture long-range dependencies in text. This innovation laid the groundwork for the development of powerful LLMs like GPT (Generative Pre-trained Transformer) and BERT (Bidirectional Encoder Representations from Transformers), both of which have since become benchmarks in AI research and applications. Google's early investments in machine learning infrastructure, coupled with their open sourcing of these technologies,

²⁴ [Defending Ukraine: Early Lessons from the Cyber War](#)

²⁵ [An overview of Russia's cyberattack activity in Ukraine](#)

²⁶ [As Tanks Rolled into Ukraine, So Did Malware. Then Microsoft entered the War.](#)

²⁷ [Starlink service is now active in Ukraine. More terminals en route.](#)

²⁸ [SpaceX shipment of Starlink satellite-internet dishes arrives in Ukraine, government official says](#)

²⁹ [How Elon Musk's satellites have saved Ukraine and changed warfare](#)

³⁰ [Will China's Reliance on Taiwanese Chips Prevent a War?](#)

³¹ [The War Over the World's Most Critical Technology: A Conversation with Chris Miller](#)

³² [Attention Is All You Need](#)

facilitated rapid advancements across the AI community, establishing the company as a leader in the ongoing AI revolution.

Microsoft's AI-driven approach significantly accelerated the development of a new battery with 70% less lithium, achieving in just 9 to 18 months what would have traditionally taken 5 to 10 years.³³ This breakthrough was made possible by integrating AI at every stage of the process, from material discovery to prototype testing. Using the Azure Quantum Elements platform, Microsoft's AI rapidly screened over 32 million potential materials, narrowing them down to a handful of viable candidates in just 80 hours—a task that would have taken years through conventional methods.³⁴ This new approach can predict material properties 1,500 times faster than traditional computational methods, allowing researchers to identify stable, high-performing materials more quickly. Once the optimal material was identified—a sodium-lithium compound—the research team at the Pacific Northwest National Laboratory (PNNL) synthesized and tested it, successfully creating a working battery prototype.³⁵ This innovative approach not only reduced lithium usage but also demonstrated the immense potential of AI in accelerating material science and technological innovation.

However, great power competition is not won by an industry that fuels the world's greatest economic system alone, much less one of the MAG7 companies alone. Microsoft Chief Executive Officer's (CEO) Satya Nadella's ambitious goal to “compress the next 250 years of chemistry and materials science progress into the next twenty-five” will have tremendous national security implications if realized.³⁶ Even if the U.S. national security institution can effectively capitalize on the advancements made by Microsoft and companies like them, we will undoubtedly fall short of the PRC's 2027 and 2035 milestones. To maintain a national competitive advantage, it is paramount to follow an AI strategic roadmap integrated across government, industry, and science.

Strategic Integration of AI into the GSI Ecosystem

The 2024 Annual Threat Assessment (ATA) published by the U.S. Intelligence Community (USIC) highlighted the need to capitalize on technological advancements through the message:

New technologies—particularly in the fields of biotechnology—are being developed and are proliferating at a rate that makes it challenging for companies and governments to shape norms regarding civil liberties, privacy, and ethics. The convergence of these emerging technologies is likely to create breakthroughs, which could lead to the rapid development of asymmetric threats—such as advanced UAVs—to U.S. interests and probably will help shape U.S. economic prosperity.³⁷

To lead in the evolving geopolitical landscape, the U.S. must excel in AI, not just in narrow or general AI separately, but in their synergistic integration across the three waves of AI: handcrafted

³³ [Scientists used AI to build a low-lithium battery from a new material that took just hours to discover](#)

³⁴ [Discoveries in weeks, not years: How AI and high-performance computing are speeding up scientific discovery](#)

³⁵ [Unlocking a new era for scientific discovery with AI: How Microsoft's AI screened over 32 million candidates to find a better battery](#)

³⁶ [Azure Quantum Elements aims to compress 250 years of chemistry into the next 25 \(microsoft.com\)](#)

³⁷ [Annual Threat Assessment of the U.S. Intelligence Community](#)

knowledge, statistical learning, and contextual adaptation.³⁸ The prevailing perception often views AI as a dichotomy; however, true innovation and the generation of new knowledge will only emerge from their collaboration. Historically, humans have been the primary agents applying cognitive brainpower to solve complex problems. Over the past five decades, there has been a concerted effort to leverage computers to alleviate some of this cognitive workload. The evolution of narrow AI, from recognizing numbers and letters for postal applications to the sophisticated computer vision algorithms pioneered by Fei-Fei Li, exemplifies the capacity of machines to perform human tasks at both human and superhuman levels.³⁹ More recently, general AI, as demonstrated by large language models, has showcased the ability to generate information at an unprecedented pace. To fully capitalize on information created by general AI, we must develop new methods that allow for the transformation of it into knowledge. The orchestration of general AI with narrow AI, use of traditional computational methods and representations, as well as encoding human workflow processes is core to unlocking the full potential. Enabling the scientific discovery process through AI is a ripe problem and one that is key to maintaining a national competitive edge.

Science is the Cornerstone of Knowledge and Innovation

The scientific method is the cornerstone of scientific discovery, and accelerating this process is pivotal for fostering faster innovation. Integrating AI with the scientific method holds immense potential in assisting scientists with their tasks. For instance, models like DeepMind’s AlphaFold3 have revolutionized fields such as materials science and biology by predicting protein structures with remarkable accuracy.⁴⁰ This generative AI revolution has sparked interest in applying AI to specific scientific domains, enhancing the efficiency and scope of research. The transformational shift brought about by agentic AI lies in its ability to orchestrate the scientific method while encoding a scientist’s values throughout the entire journey.⁴¹ Unlike narrow AI models such as AlphaFold3 and ESMFold, which fall under the second wave of the Defense Advance Research Project Agency’s (DARPA) AI roadmap and are limited to specific tasks within the hypothesis-to-experiment phase, agentic AI encompasses a broader spectrum.^{42,43} However, traditional AI models are still incredibly important and must be used in coordination with generative AI. For example, Azure Quantum Elements exemplifies how new deep learning models can be trained to speed up structural relaxation by over 1,500 times, going from 78 central processing unit (CPU) hours per structural relaxation to a little less than 3 CPU minutes with new AI innovations.⁴⁴ With agentic AI, we will be able to extend across the entire scientific method to autonomously generate hypotheses, design experiments, analyze data, and even draw conclusions, thereby accelerating the

³⁸ [DARPA Perspective on AI](#)

³⁹ [Searching for Computer Vision North Stars](#)

⁴⁰ [Accurate structure prediction of biomolecular interactions with AlphaFold 3](#)

⁴¹ Agentic AI refers the ability for multiple LLMs or generative models to work individually or collectively to solve a goal. An individual instance of a generative model is referred to as an “agent”. Each agent often has the ability to plan domain-specific tools to accomplish a task. For agentic AI to be faithful and trustworthy, humans must not be disenfranchised from the process; they must work alongside the agents. It is a symbiotic system between AI agents and humans working as an integrated team to achieve a common goal defined by humans.

⁴² [Accurate structure prediction of biomolecular interactions with AlphaFold 3](#)

⁴³ [Evolutionary-scale prediction of atomic-level protein structure with a language model](#)

⁴⁴ [Accelerating computational materials discovery with artificial intelligence and cloud high-performance computing: from large-scale screening to experimental validation](#)

entire scientific process.⁴⁵ Accelerating the scientific method with agentic AI is not just a technological advancement; it is transformational for society as a whole. The transformation potential that it has bears a great cost if misused. Thus, what is equally crucial here is the encoding of appropriate values into AI and scientific discovery.

For the government, access to hyperscale AI is merely the beginning; it should not be the end state. Today's focus must extend beyond developing trustworthy AI, addressing challenges such as hallucinations, to encompass the broader goal of embedding ethical principles and values into AI systems.^{46,47} While critical, trustworthy AI must naturally be built into agentic AI systems and are an added benefit of using a network of AI models. By ensuring that domain-specific tasks are delegated to the respective domain's model, we raise trust in the system; engineering principles and design choices feed into the trustworthiness of agentic AI systems and outputs. As agentic AI is applied to the scientific method, we must ensure that scientists are brought along in the journey, using their own knowledge and school of thought, tools of choice, and the ability to raise flags when necessary. This holistic approach ensures that AI-driven scientific advancements that contribute to U.S. and allied interests.

Technology Feedback Cycle in Sustaining Integrated Deterrence

Understanding the ecosystem between government, science, and industry and the intricacies of even one vertical alone is challenging enough. Our aim here is not to talk about all the instruments within each that can be exercised to reduce or mitigate the influence of cross-vertical events, rather to show how the U.S. may be susceptible to seemingly harmless shocks that have massive effects. We examine this by providing positive reinforcing technology loops as well as negative ones that may be observed. The government, science, and industry are fundamentally linked to one another and to many others; separation of one pillar can have catastrophic effects on the U.S.'s full operating potential.

The development of Google's Transformer technology has set off a chain reaction of influence that reverberates through science, industry, and government, creating a circular loop of innovation and policy adaptation. National laboratories and academic institutions have adopted this technology to enhance their research capabilities, leading to significant advancements in various scientific fields. For instance, improved data analysis techniques have accelerated discoveries in genomics, climate modeling, and materials science. These scientific breakthroughs, in turn, drive further innovation in the tech industry, leading to the development of more advanced AI applications and tools. Recognizing the transformative potential of AI technologies, the U.S. government has responded with policies aimed at supporting and regulating AI development. The (Creating Helpful Incentives to Produce Semiconductors) CHIPS Act and Inflation Reduction Act (IRA), for example, were enacted to bolster domestic semiconductor manufacturing and research, acknowledging the strategic importance of AI and advanced computing.^{48,49} This legislation provides substantial

⁴⁵ [Augmenting large language models with chemistry tools](#)

⁴⁶ [Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence | The White House](#)

⁴⁷ [Supercharging Research: Harnessing Artificial Intelligence to Meet Global Challenges](#)

⁴⁸ [H.R.4346 - CHIPS and Science Act](#)

⁴⁹ [H.R.5376 - Inflation Reduction Act of 2022](#)

funding for semiconductor research and development, ensuring that the U.S. remains competitive in the global technology landscape. Additionally, critical AI regulation initiatives, such as the Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, highlight the U.S. government's commitment to harnessing AI's benefits while mitigating its risks.⁵⁰ These regulations aim to ensure that AI technologies are developed and deployed responsibly, addressing security, privacy, and ethical concerns. The interplay between Transformer's development, scientific research, and government policy creates a positive reinforcing technology loop. Advances in AI technology drive scientific innovation, which informs and shapes government policies. These policies, such as the CHIPS Act and AI regulations, provide the necessary support and framework for further technological advancements, creating a cycle of continuous improvement and innovation. In this circular loop vision, the development of Transformer's technology influences government policies downstream, which in turn, create an environment conducive to further AI advancements. This cycle ensures that the U.S. remains at the forefront of technological innovation, enhancing its strategic position and integrated deterrence capabilities. By understanding and leveraging these positive feedback cycles, the U.S. can maintain resilience and adaptability in the face of emerging technological challenges. This positive reinforcing loop is beneficial because it bolsters the U.S.'s ability to deter and protect high-value national interests. By continuously advancing AI technology and adapting policies to support these advancements, the U.S. strengthens its strategic position, ensuring it can effectively respond to and mitigate potential threats. This integrated approach enhances national security, economic stability, and technological leadership, safeguarding key U.S. interests.

Conversely, negative feedback cycles can occur when a particular segment of the GSI ecosystem is under strain. These strains may be induced by the misuse or mismanagement of finite natural resources like water, one that is critical in cooling datacenters, chip manufacturing, and other industrial processes. Water is increasingly subject to fluctuations of natural events like hurricanes or droughts, where water shortages will directly impact the cooling of datacenters and subsequently reduce computational power and efficiency. This bottleneck at the bottom of the AI stack will cascade negatively to the very top of the stack: AI applications, fundamentally slowing down scientific research and development. For example, in regions like Arizona, where natural water sources are already limited, the impact of droughts on data centers is particularly pronounced.⁵¹ The impacts of negative feedback cycles will be a detriment to U.S. national security interests; enabling a positive cycle is key to continually winning integrated deterrence.

Strategic Roadmap for an AI-enabled Ecosystem

U.S. Deputy Secretary of Defense Robert Work's development of the DoD's third offset strategy highlighted the critical importance of adaptability to maintain hegemony in an era where China's military modernization and expansion pose a significant threat.^{52,53} This shift did not imply an immediate pivot away from persistent threats in the Middle East, rather it emphasized the need for the U.S. to possess the ability to address multiple, evolving challenges simultaneously through a strategic reallocation of resources. Secretary Work's principle holds especially true today, as the

⁵⁰ [Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence](#)

⁵¹ [Monitoring Drought in Arizona](#)

⁵² [Reagan National Defense Forum Keynote, Chuck Hagel](#)

⁵³ [U.S. Department of Defense, Military and Security Developments Involving the People's Republic of China, 2023](#)

U.S. must strategically reallocate priorities and resources to foster an AI-enabled ecosystem that enhances our strategic edge by making us faster, more innovative, and more resilient in our operations and infrastructure.

The PRC aims to build a “great modern socialist society”, it faces a multitude of challenges in resolving a fundamental contradiction – a clash of ideological systems – primarily against the United States.⁵⁴ The PRC has demonstrated its ability to leverage the advantages of an autocratic, socialist nature. Through the Chinese Communist Party’s Central Commission, the Chinese government controls the direction of scientific endeavors such as the 7 Sons of National Defense and critical research institutions, as well as industry giants like Baidu and Tencent, which are all subject to the Party’s direction through the Ministry of Industry and Information Technology.^{55,56,57} Fundamentally, their advantage lies in a concentrated approach, leveraging high amounts of cognitive power derived through a “law of large numbers” strategy.⁵⁸

The PRC's ability to leverage its competitive edge is the fundamental reason why the U.S. has to have an “*AI Roadmap to Winning Great Power Competition*” with the goal of transforming knowledge into a perpetual engine that drives America’s strategic advantage in the decades to come. Without this comprehensive AI Strategy, the U.S. will squander its existing innovative potential, leading to an ineffective translation of these advancements to national security priorities by the time the PRC achieves its goals. Therefore, the U.S. must have an intertwined roadmap across government, science, and industry that spans the entire AI stack – from applications and models to infrastructure, chips, energy, and winning global mindshare.

Government Focus Areas

The government's role in shaping an AI-enabled future is crucial for maintaining the United States' strategic advantage in an era of rapid technological advancement and great power competition. By acting as both a facilitator of scientific discovery and an integrator of industry products, the government can create an environment that not only fosters innovation but also ensures that AI technologies are developed and integrated in ways that align with national security interests. In this section, we outline key focus areas where government action is essential to harness the full potential of AI across various sectors. While we provide an overview of these strategic initiatives here, we will delve deeper into each area in future government-focused articles to offer greater insights and actionable recommendations.

1. **Policy Innovation and Support:** The U.S. government should prioritize creating a dynamic regulatory framework that encourages AI integration while protecting national security interests. These policies should focus on understanding incentive structures in public-private partnerships, ensuring that industry and scientific sectors can rapidly develop and scale AI technologies. Furthermore, the government must also craft targeted legislation that reshapes

⁵⁴ IV, [U.S. Department of Defense, Military and Security Developments Involving the People’s Republic of China, 2023](#)

⁵⁵ [The Chinese Defence Universities Tracker](#)

⁵⁶ [Universities and the Chinese Defense Technology Workforce](#)

⁵⁷ Page 23, [Hearing on Technology, Trade, and Military-Civil Fusion: China’s Pursuit of Artificial Intelligence, New Materials, and New Energy](#)

⁵⁸ [China’s Alternative Order | Foreign Affairs](#)

the landscape of strategic technology development to safeguard interests. This may include legislation that targets continuing to reshore critical technologies (to the U.S. or global allies) and expanding innovation partnerships with key global allies.

2. **Fostering government understanding of large tech companies:** The U.S. government must play an active role in fostering an AI-enabled ecosystem by supporting both large enterprises and small and medium-sized businesses (SMBs). There needs to be new programs to develop tighter strategic and commercial alignment between MAG7 companies and the government. Programs focused on developing leaders who can gain hands-on experiences with the advanced tech industry's business, research and development, and acquisition processes to enhance the U.S. Department of Defense's (DoD) technology adoption strategy and processes. Understanding incentives of large businesses building enterprise products and how to best incentivize industry stakeholders to focus on minimizing bespoke use-cases of enterprise technology.
3. **National AI Strategy with Specific Focus Areas:** The U.S. government should develop a more granular AI strategy that outlines specific focus areas for technology integration and development. This would include subfields such as AI for energy independence, advanced defense systems, supply chain security, and infrastructure resilience. Tailoring policies to these key areas will ensure that the nation's AI efforts are aligned with both economic competitiveness and national security priorities.
4. **Intelligence Integration into Industry:** Greater collaborative efforts between U.S. government and industry, with the government sharing intelligence to safeguard integral technologies being built by industry. This integration would provide comprehensive situational awareness, enabling decision-makers to make more informed product decisions and understand the cybersecurity risk to their systems.
5. **AI-Powered Defense Modernization:** The U.S. government should accelerate efforts to integrate AI across all branches of the military and defense infrastructure as an industry fast follower. AI can be utilized to enhance command, control, communication, and intelligence (C4ISR) systems, improving real-time situational awareness and decision-making.
6. **Leveraging AI Industry Infrastructure:** To keep pace with private sector innovation, the focus should be on U.S. government and industry partnerships to leverage the industry investment in AI infrastructure required for advanced AI research. This includes high-performance computing resources, data storage solutions, and AI model training capabilities.
7. **Strategic AI Alliances and Partnerships:** The U.S. government must prioritize alliances with key strategic partners, particularly in Europe and the Indo-Pacific, to create a coalition of democracies that lead in AI innovation. The U.S. and its allies can maintain a technological edge over adversaries like the PRC and Russia by jointly developing AI standards, sharing research, and coordinating defense strategies. These alliances should extend beyond traditional

cooperation and focus on creating an enterprise ecosystem that reduces the procurement of country-specific bespoke technologies.

Industry Alignment

The government alignment between industry initiatives and national security objectives are essential, particularly as we face accelerating advancements from global competitors like the PRC. In this section, we outline key areas where industry collaboration and proactive engagement are vital to fortifying the nation's AI capabilities. While we provide an overview of these strategic priorities here, we will delve deeper into each focus area in future industry-centric articles to offer great insights and actionable recommendations.

1. **Securing Critical AI Supply Chains:** Industry must take proactive steps to secure AI technology supply chains, particularly in areas like semiconductor manufacturing, rare earth minerals, and high-performance computing infrastructure (i.e., energy, water). As an economic engine of technological advancement is critical for maintaining a U.S. advantage, the industry holds a crucial responsibility to sustain the United States' leadership in AI and maintain its strategic advantage on the global stage. security and stability of AI-related supply chains will be critical in protecting national interests from disruptions caused by adversarial actions.
2. **Investing in AI Security and Resilience:** Industry must take proactive measures to secure AI digital infrastructure from foreign adversaries. The MAG7 Companies, which have a significant global presence, should continue investing in secure cloud development practices and AI security platforms to safeguard U.S. interests. Partnering with the U.S. and allies to focus on security and resilience in the face of geopolitical threats is key to maintaining long-term technological leadership.
3. **AI to Enhance Supply Chain Resilience:** Industry leaders, particularly in manufacturing, logistics, and technology sectors, should invest in integrating AI solutions that enhance global supply chain resiliency. AI should increasingly be leveraged to identify and mitigate supply chain vulnerabilities, improve resource allocation, and ensure continuity in the event of geopolitical disruptions or other ecosystem shocks. Enhancing resiliency is crucial in maintaining the U.S.' operational stability and reducing dependencies on hostile or unstable regions.
4. **Dual-Use AI Technology Development:** The U.S. government should work closely with Industry to drive requirements that enable dual-use capabilities, those with both civilian and military applications, from industry's development of AI technologies. This ensures that advancements made in commercial AI (i.e., autonomous systems, advanced information analytics, or cybersecurity tools) can be rapidly adapted for defense purposes. Government should explore ways to bridge the gap between commercial innovation and national security needs, ensuring that the military can quickly adopt and deploy cutting-edge technologies.

5. **AI-Enabled Cybersecurity:** To counter increasingly sophisticated cyber threats, industry should lead the development of AI-based cybersecurity systems. These systems should be designed to detect, predict, and neutralize cyberattacks in real time. By integrating AI into the nation's and industry's critical infrastructure defense mechanisms, both can protect against nation-state actors seeking to undermine U.S. national security through cyber warfare and espionage.

Fueling Scientific Advancement

As the cornerstone of the driving force behind technological breakthroughs, scientific advancement is essential for sustaining the United States' competitive edge in the global arena. The integration of Artificial Intelligence into scientific research accelerates discovery, enabling transformative developments that can bolster national security and economic prosperity. In this section, we highlight key strategies for fueling scientific advancement through AI, emphasizing the pivotal role of collaboration among government, industry, and the scientific community. While we provide an overview of how to fuel scientific advancement, we will delve deeper into each focus area in future scientific-focused articles to offer greater insights and actionable recommendations.

1. **AI for Fundamental Research:** Scientists should leverage AI to accelerate breakthroughs in fundamental research. AI can assist in analyzing large datasets, simulating complex systems, and predicting outcomes in fields like quantum mechanics, space exploration, and materials science. By incorporating AI-driven tools into fundamental research, scientists can more quickly generate hypotheses, test theories, and make discoveries that have far-reaching implications.
2. **Accelerating Scientific Discovery with AI:** The U.S. scientific community must continue integrating AI into accelerating the scientific method. Scientists should work toward expanding the use of generative AI and agentic AI in fields like biotechnology, materials science, and energy, speeding up the hypothesis-experiment cycle and leading to faster, more impactful discoveries. Agentic AI should extend across the scientific method to autonomously generate hypotheses, design experiments, analyze data, and synthesize materials in wet lab environments, thereby accelerating new knowledge creation.
3. **Investment in AI Integration into Foundational Science:** Federal investments are necessary to develop and deploy AI systems aligning with national security objectives. Strategic funding should focus on high-risk, high-reward research areas to maintain U.S. innovation and technological dominance. This includes funding for quantum computing, advanced materials, and biotechnologies, where AI can serve as a driving force in accelerating breakthroughs and benefit from scientific advancements to fuel the next AI wave of discovery.
4. **Collaboration with Government and Industry:** Scientific institutions should focus on working in partnership with both the government and industry in developing technologies that enhance national competitiveness by fostering interdisciplinary AI research that integrates

fields like biology, physics, social sciences, and engineering. The focus should be on efforts that both affect national security and win global mind share; science can drive holistic, impactful innovations by promoting interdisciplinary research. Research should be focused on complex, such as complex cross-cutting problems, such as disease prevention, climate change, and sustainable energy solutions, to expand U.S. global mindshare dominance.

Conclusion

The U.S. has fostered the greatest technology development ecosystem in the world, fueling ongoing scientific innovation and entrepreneurial success. This is evidenced by the fact that the MAG7 are all U.S.-based companies, the majority of which did not even exist thirty years ago and now have a combined market capitalization of \$15 trillion.⁵⁹ Industry has led the way in AI technology development and integration, with single MAG7 companies making investments in AI development and infrastructure greater than most governments make in their national defense.⁶⁰ As highlighted in 2024 Commission on the National Defense Strategy for the United States, “U.S. military lacks both the capabilities and the capacity required to be confident it can deter and prevail in combat. It needs to do a better job of incorporating new technology at scale; field more and higher-capability platforms, software, and munitions; and deploy innovative operational concepts to employ them together better. The war in Ukraine has demonstrated the need to prepare for new forms of conflict and to integrate technology and new capabilities rapidly with older systems. Such technologies include swarms of attritable systems, artificial intelligence-enabled capabilities, hypersonics and electronic warfare, fully integrated cyber and space capabilities, and vigorous competition in the information domain.”⁶¹

With 2027 only three years away, and U.S. defense’s “innovation adoption problem,”⁶² we will be even further behind if there is not a well-defined strategy that enables seamless integration of the industry’s investment in AI with a government focused on driving industry alignment and fueling scientific advancement. While these goals are ambitious, defining this strategy requires us first to acknowledge the ongoing shift in the global order and take decisive action to bolster U.S. Artificial Intelligence (AI) capabilities and strengthen the U.S. competitive edge. The U.S. needs an integrated approach to ensure that we can effectively respond to the PRC’s 2027 and 2035 milestones or risk losing U.S. hegemony. The U.S. stands at a pivotal moment in great power competition, where AI plays a decisive role in shaping the future of national security. Now more than ever, following an AI strategic roadmap integrated across government, science, and industry is paramount.

⁵⁹ Record breaking: 'Magnificent Seven' stocks exceed \$15 trillion market cap for the first time (finbold.com)

60 [Military spending by country worldwide 2023 | Statista](#)

61 Commission on the National Defense Strategy - 2024

⁶² Statement of Dr. Eric Schmidt House Armed Services Committee April 17, 2018